ValPay Portfolio 4: NoSQL (MongoDB)

MIT 8103 Advanced Database Systems · Evidence

Valentine Efagene

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Compiled from `portfolio-4-nosql/evidence`: 19 screenshots in `screenshots/` and script output in `script-output/` (`cross_store_check.txt`, `explain_plans.txt`, `load_summary.txt`, `query_results.txt`, `validation_rejection.txt`). Text output is reproduced verbatim.

1. SEED LOAD

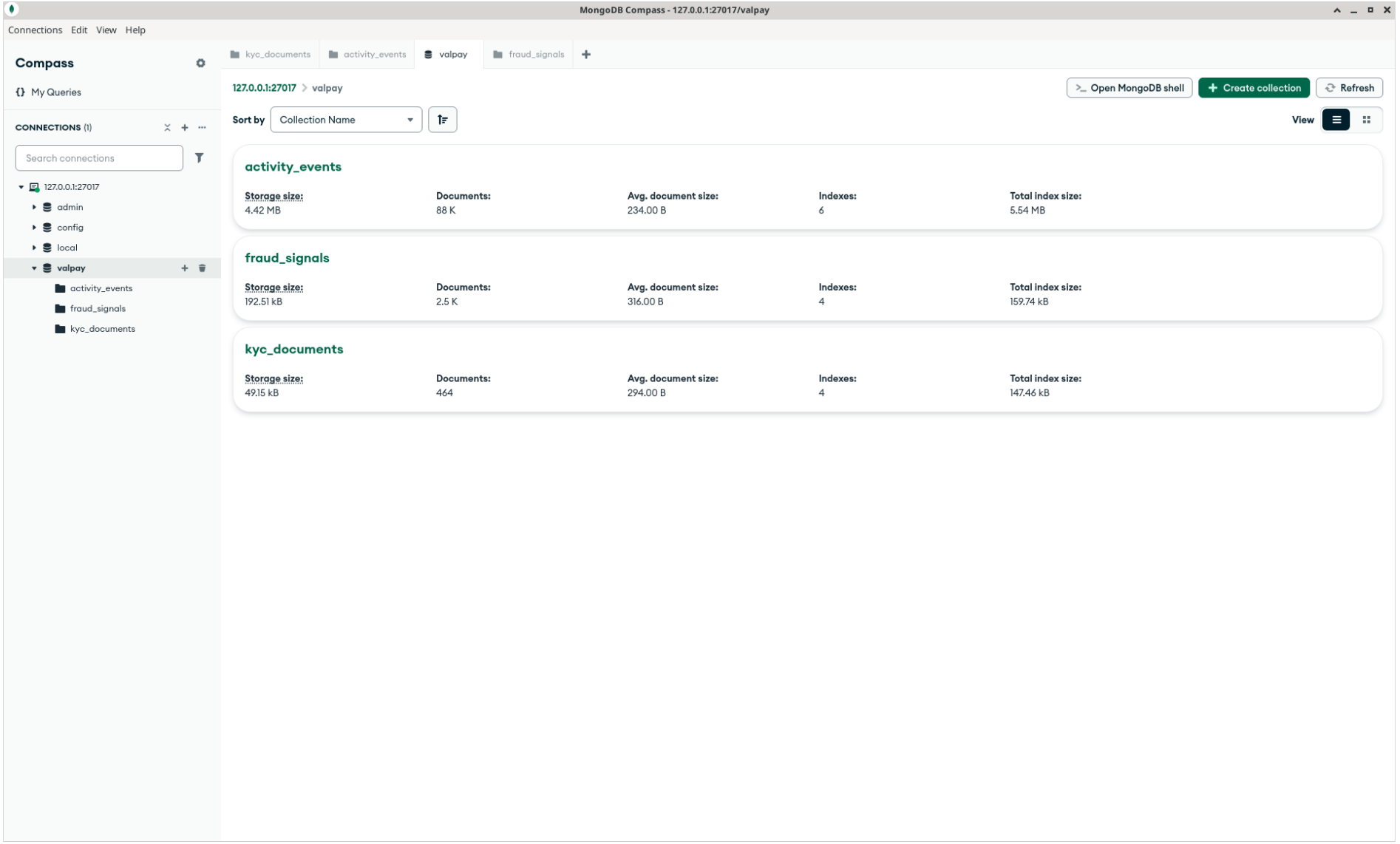


Figure 1. *MongoDB Compass: collections in the valpay database with document counts.*

```
portfolio-4-nosql $ mongosh --quiet valpay --file seed.mongodb.js | head -39
=== ValPay MongoDB seed (database: valpay, MongoDB 7.0.25) ===
Customers covered: 202 (customer_id 1, 2 and 4-203; 3 is the ops wallet)

kyc_documents       464 documents
activity_events   87886 documents
fraud_signals      2484 documents

activity_events by event:
  login_success         28003
  otp_sent              28003
  transfer_initiated    28003
  login_failed           2297
  fraud_rule_hit         1580
  time span: 2025-01-01T00:08:59.000Z to 2026-10-31T23:49:14.000Z

kyc_documents by status:
  PENDING                  44
  REJECTED                 46
  VERIFIED                374

fraud_signals by signal_type:
  DEVICE_SUMMARY           904
  RULE_HIT                1580

Indexes:
  kyc_documents._id_ {"_id":1}
  kyc_documents.customer_doc_latest {"customer_id":1,"document_type":1,"submitted_at":-1}
  kyc_documents.pending_review_queue {"submitted_at":1}  partial {"status":"PENDING"}
  kyc_documents.status_doc_type {"status":1,"document_type":1}
  activity_events._id_ {"_id":1}
  activity_events.customer_timeline {"customer_id":1,"ts":-1}
  activity_events.event_time {"event":1,"ts":-1}
  activity_events.device_time {"device_id":1,"ts":-1}
  activity_events.txn_reference {"reference":1}  sparse
  activity_events.retention_ttl_3y {"ts":1}  TTL 94608000s
  fraud_signals._id_ {"_id":1}
  fraud_signals.type_risk {"signal_type":1,"risk_score":-1}
  fraud_signals.case_queue {"signal_type":1,"case_status":1,"detected_at":-1}
  fraud_signals.customer_cases {"customer_id":1,"detected_at":-1}  partial {"signal_type":"RULE_HIT"}
portfolio-4-nosql $
```

Figure 2. *mongosh: seed load summary with document counts, breakdowns and indexes.*

Source: `evidence/script-output/load_summary.txt`

```
=== ValPay MongoDB seed (database: valpay, MongoDB 7.0.45) ===
Customers covered: 202 (customer_id 1, 2 and 4-203; 3 is the ops wallet)

kyc_documents       464 documents
activity_events   87886 documents
fraud_signals      2484 documents

activity_events by event:
  login_success         28003
  otp_sent              28003
  transfer_initiated    28003
  login_failed           2297
  fraud_rule_hit         1580
  time span: 2025-01-01T00:08:59.000Z to 2026-10-31T23:49:14.000Z

kyc_documents by status:
  PENDING                  44
  REJECTED                 46
  VERIFIED                374

fraud_signals by signal_type:
  DEVICE_SUMMARY           904
  RULE_HIT                1580

Indexes:
```

```
  kyc_documents._id_ {"_id":1}
  kyc_documents.customer_doc_latest {"customer_id":1,"document_type":1,"submitted_at":-1}
  kyc_documents.pending_review_queue {"submitted_at":1}  partial {"status":"PENDING"}
  kyc_documents.status_doc_type {"status":1,"document_type":1}
  activity_events._id_ {"_id":1}
  activity_events.customer_timeline {"customer_id":1,"ts":-1}
  activity_events.event_time {"event":1,"ts":-1}
  activity_events.device_time {"device_id":1,"ts":-1}
  activity_events.txn_reference {"reference":1}  sparse
  activity_events.retention_ttl_3y {"ts":1}  TTL 94608000s
  fraud_signals._id_ {"_id":1}
  fraud_signals.type_risk {"signal_type":1,"risk_score":-1}
  fraud_signals.case_queue {"signal_type":1,"case_status":1,"detected_at":-1}
  fraud_signals.customer_cases {"customer_id":1,"detected_at":-1}  partial
{"signal_type":"RULE_HIT"}

kyc_documents validator (from kyc_schema.json plus the REJECTED rule):
{
  validator: {
    '$and': [
      {
        '$jsonSchema': {
          bsonType: 'object',
          required: [
            'customer_id',
            'account_number',
            'document_type',
            'status',
            'metadata',
            'submitted_at'
          ],
          properties: {
            customer_id: {
              bsonType: 'int',
              minimum: 1,
              description: 'customer.customer_id in PostgreSQL (Portfolio 1)'
            },
            account_number: {
              bsonType: 'string',
              pattern: '^VP[0-9]{10}$',
              description: 'wallet_account.account_number in PostgreSQL'
            },
            document_type: {
              enum: [
                'NIN',
                'BVN',
                'PASSPORT',
                'UTILITY_BILL'
              ]
            },
            status: {
              enum: [
                'PENDING',
                'VERIFIED',
                'REJECTED'
              ]
            },
            metadata: {
              bsonType: 'object',
              description: 'Fields vary by document_type (NIN, BVN, passport, utility bill)'
            },
            submitted_at: {
              bsonType: 'date'
            },
            reviewed_at: {
              bsonType: 'date'
            },
            reviewer: {
              bsonType: 'string'
```

```
            }
          }
        }
      },
      {
        '$or': [
          {
            status: {
              '$ne': 'REJECTED'
            }
          },
          {
            'metadata.rejection_reason': {
              '$type': 'string'
            }
          }
        ]
      }
    ]
  },
  validationLevel: 'strict',
  validationAction: 'error'
}
```

2. INDEX EVIDENCE (EXPLAIN)

Customer timeline and transfer window

```
portfolio-4-nosql $ mongosh --quiet valpay --file explain.mongodb.js | awk '/=== E1/,/=== E3/' | grep -v '=== E3'
=== E1. Customer timeline: customer_id 57, latest 20 events ===
Collection: activity_events (87886 documents)
Index under test: customer_timeline {"customer_id":1,"ts":-1}
  BEFORE (only _id index)
    winning plan       : SORT <- COLLSCAN
    nReturned          : 20
    totalKeysExamined  : 0
    totalDocsExamined  : 87886
    executionTimeMillis: 27
  AFTER (customer_timeline created)
    winning plan       : LIMIT <- FETCH <- IXSCAN(customer_timeline)
    nReturned          : 20
    totalKeysExamined  : 20
    totalDocsExamined  : 20
    executionTimeMillis: 0
  Docs examined: 87886 -> 20 (4394x fewer)

=== E2. transfer_initiated events, 1-7 October 2026 ===
Collection: activity_events (87886 documents)
Index under test: event_time {"event":1,"ts":-1}
  BEFORE (only _id index)
    winning plan       : COLLSCAN
    nReturned          : 293
    totalKeysExamined  : 0
    totalDocsExamined  : 87886
    executionTimeMillis: 27
  AFTER (event_time created)
    winning plan       : FETCH <- IXSCAN(event_time)
    nReturned          : 293
    totalKeysExamined  : 293
    totalDocsExamined  : 293
    executionTimeMillis: 0
  Docs examined: 87886 -> 293 (300x fewer)

portfolio-4-nosql $
```

Figure 3. *mongosh: explain() for the customer timeline and the transfer_initiated window, before and after indexing.*

Source: `evidence/script-output/explain_plans.txt`

```
=== E1. Customer timeline: customer_id 57, latest 20 events ===
Collection: activity_events (87886 documents)
Index under test: customer_timeline {"customer_id":1,"ts":-1}
  BEFORE (only _id index)
    winning plan       : SORT <- COLLSCAN
    nReturned          : 20
    totalKeysExamined  : 0
    totalDocsExamined  : 87886
    executionTimeMillis: 19
  AFTER (customer_timeline created)
    winning plan       : LIMIT <- FETCH <- IXSCAN(customer_timeline)
    nReturned          : 20
    totalKeysExamined  : 20
    totalDocsExamined  : 20
    executionTimeMillis: 0
  Docs examined: 87886 -> 20 (4394x fewer)
```

Source: `evidence/script-output/explain_plans.txt`

```
=== E2. transfer_initiated events, 1-7 October 2026 ===
Collection: activity_events (87886 documents)
Index under test: event_time {"event":1,"ts":-1}
  BEFORE (only _id index)
    winning plan       : COLLSCAN
    nReturned          : 293
    totalKeysExamined  : 0
    totalDocsExamined  : 87886
    executionTimeMillis: 27
```

```
AFTER (event_time created)
  winning plan       : FETCH <- IXSCAN(event_time)
  nReturned          : 293
  totalKeysExamined  : 293
  totalDocsExamined  : 293
  executionTimeMillis: 0
Docs examined: 87886 -> 293 (300x fewer)
```

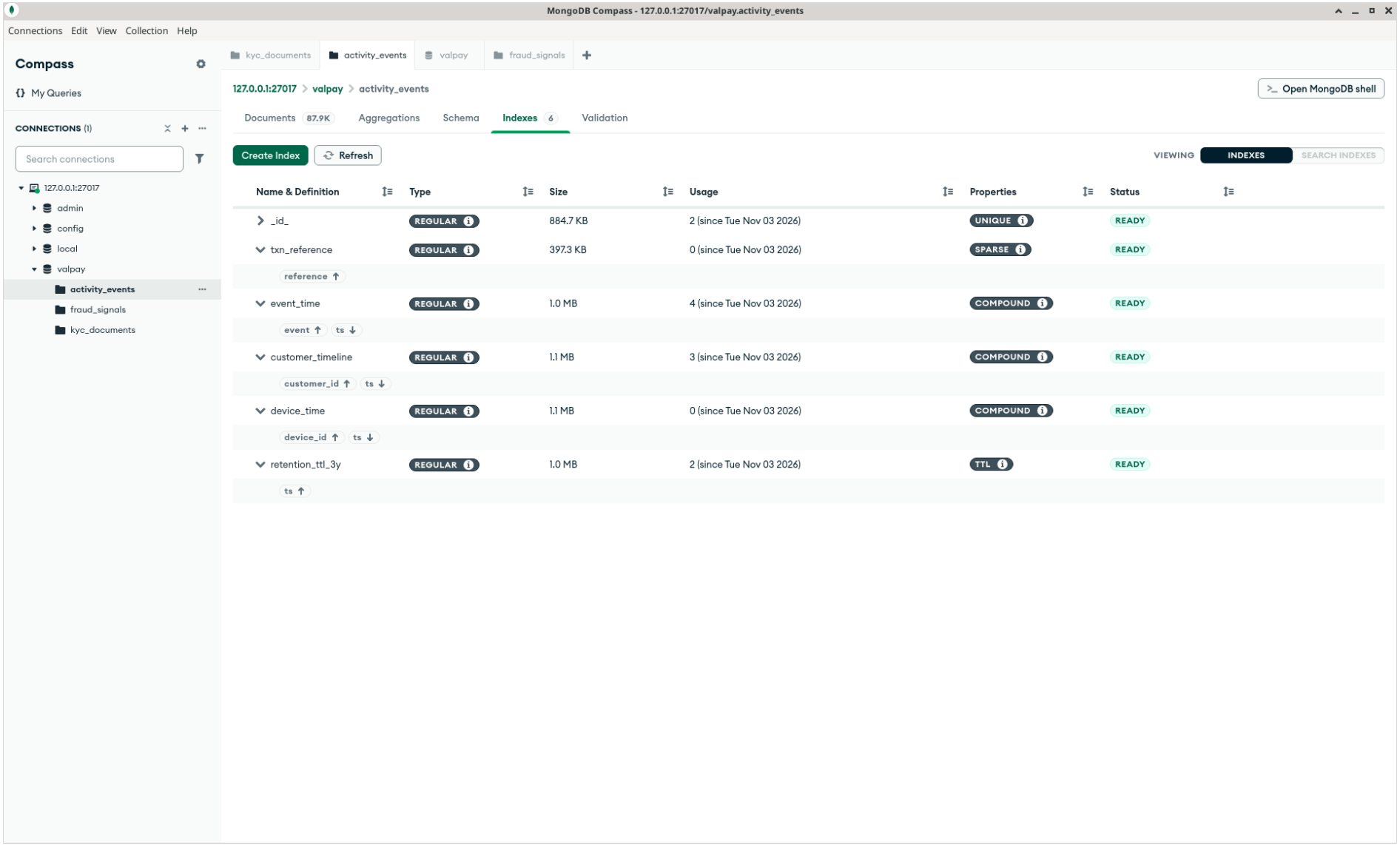


Figure 4. *MongoDB Compass: indexes on activity_events.*

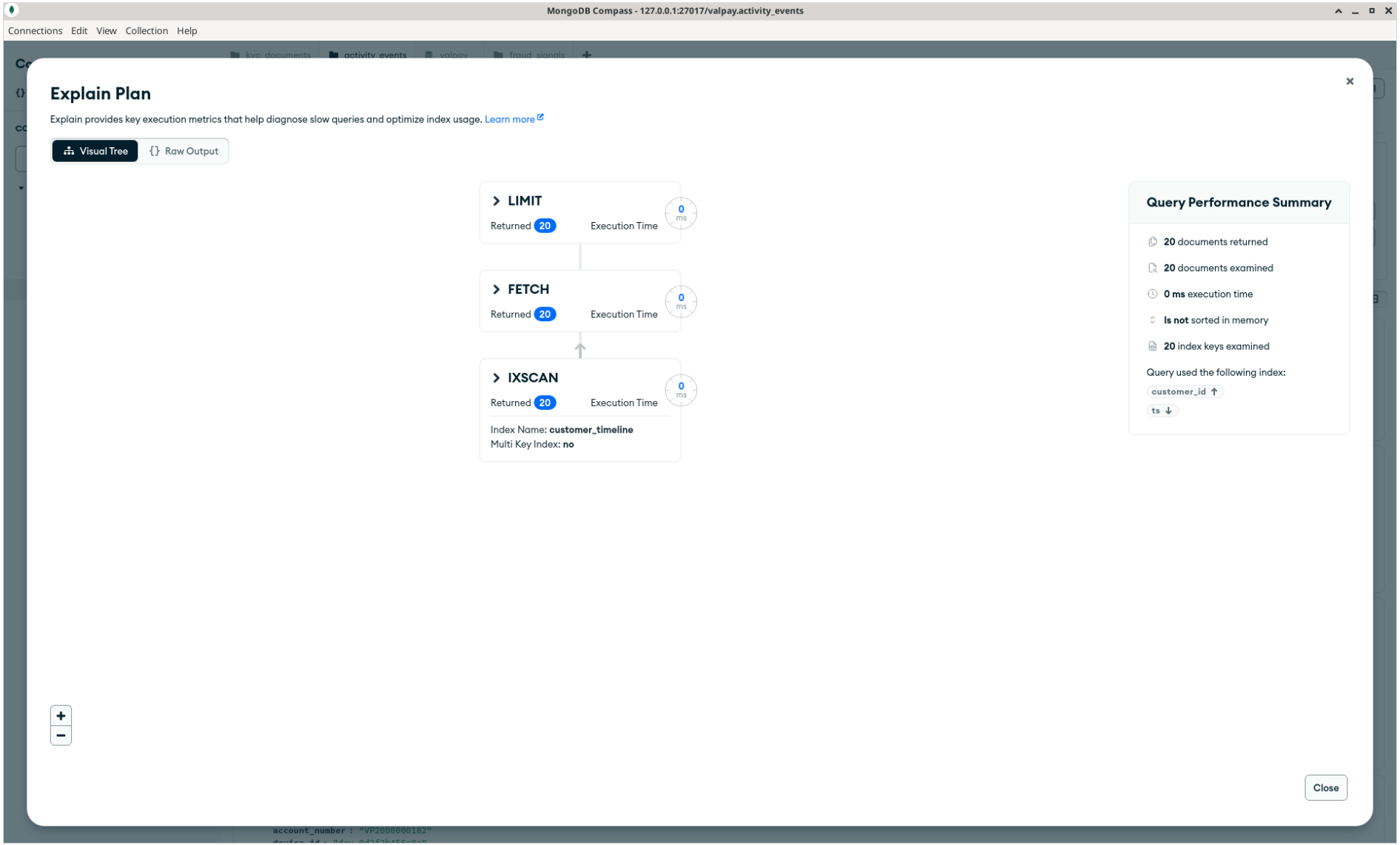


Figure 5. *MongoDB Compass: explain plan for the customer timeline query using an index scan.*

Reference trace and pending KYC queue

```
portfolio-4-nosql $ mongosh --quiet valpay --file explain.mongodb.js | awk '/=== E3/,0'
=== E3. Trace reference VP-TXN-20261026-039684 ===
Collection: activity_events (87886 documents)
Index under test: txn_reference {"reference":1} sparse
  BEFORE (only _id index)
    winning plan       : COLLSCAN
    nReturned          : 1
    totalKeysExamined  : 0
    totalDocsExamined  : 87886
    executionTimeMillis: 25
  AFTER (txn_reference created)
    winning plan       : FETCH <- IXSCAN(txn_reference)
    nReturned          : 1
    totalKeysExamined  : 1
    totalDocsExamined  : 1
    executionTimeMillis: 0
  Docs examined: 87886 -> 1 (87886x fewer)

=== E4. KYC review queue: PENDING, oldest first ===
Collection: kyc_documents (464 documents)
Index under test: pending_review_queue {"submitted_at":1} partial {"status":"PENDING"}
  BEFORE (only _id index)
    winning plan       : SORT <- COLLSCAN
    nReturned          : 44
    totalKeysExamined  : 0
    totalDocsExamined  : 464
    executionTimeMillis: 0
  AFTER (pending_review_queue created)
    winning plan       : FETCH <- IXSCAN(pending_review_queue)
    nReturned          : 44
    totalKeysExamined  : 44
    totalDocsExamined  : 44
    executionTimeMillis: 0
  Docs examined: 464 -> 44 (11x fewer)

=== Partial index coverage ===
  pending_review_queue indexes 44 of 464 kyc_documents (status PENDING only)

=== The partial index is not used for VERIFIED documents ===
  status VERIFIED sorted by submitted_at
    winning plan       : SORT <- FETCH <- IXSCAN(status_doc_type)
    nReturned          : 374
    totalKeysExamined  : 374
    totalDocsExamined  : 374
    executionTimeMillis: 0
portfolio-4-nosql $
```

Figure 6. *mongosh: explain() for the reference trace and the pending KYC queue with its partial index.*

Source: `evidence/script-output/explain_plans.txt`


```
=== E3. Trace reference VP-TXN-20261026-039684 ===
Collection: activity_events (87886 documents)
Index under test: txn_reference {"reference":1} sparse
  BEFORE (only _id index)
    winning plan       : COLLSCAN
    nReturned          : 1
    totalKeysExamined  : 0
    totalDocsExamined  : 87886
    executionTimeMillis: 25
  AFTER (txn_reference created)
    winning plan       : FETCH <- IXSCAN(txn_reference)
    nReturned          : 1
    totalKeysExamined  : 1
    totalDocsExamined  : 1
    executionTimeMillis: 0
  Docs examined: 87886 -> 1 (87886x fewer)
```

Source: evidence/script-output/explain_plans.txt

```
=== E4. KYC review queue: PENDING, oldest first ===
Collection: kyc_documents (464 documents)
Index under test: pending_review_queue {"submitted_at":1} partial {"status":"PENDING"}
  BEFORE (only _id index)
    winning plan       : SORT <- COLLSCAN
    nReturned          : 44
    totalKeysExamined  : 0
    totalDocsExamined  : 464
    executionTimeMillis: 0
  AFTER (pending_review_queue created)
    winning plan       : FETCH <- IXSCAN(pending_review_queue)
    nReturned          : 44
    totalKeysExamined  : 44
    totalDocsExamined  : 44
    executionTimeMillis: 0
  Docs examined: 464 -> 44 (11x fewer)
```

Source: evidence/script-output/explain_plans.txt

```
=== Partial index coverage ===
  pending_review_queue indexes 44 of 464 kyc_documents (status PENDING only)
```

Source: evidence/script-output/explain_plans.txt

```
=== The partial index is not used for VERIFIED documents ===
  status VERIFIED sorted by submitted_at
    winning plan       : SORT <- FETCH <- IXSCAN(status_doc_type)
    nReturned          : 374
    totalKeysExamined  : 374
    totalDocsExamined  : 374
    executionTimeMillis: 0
```

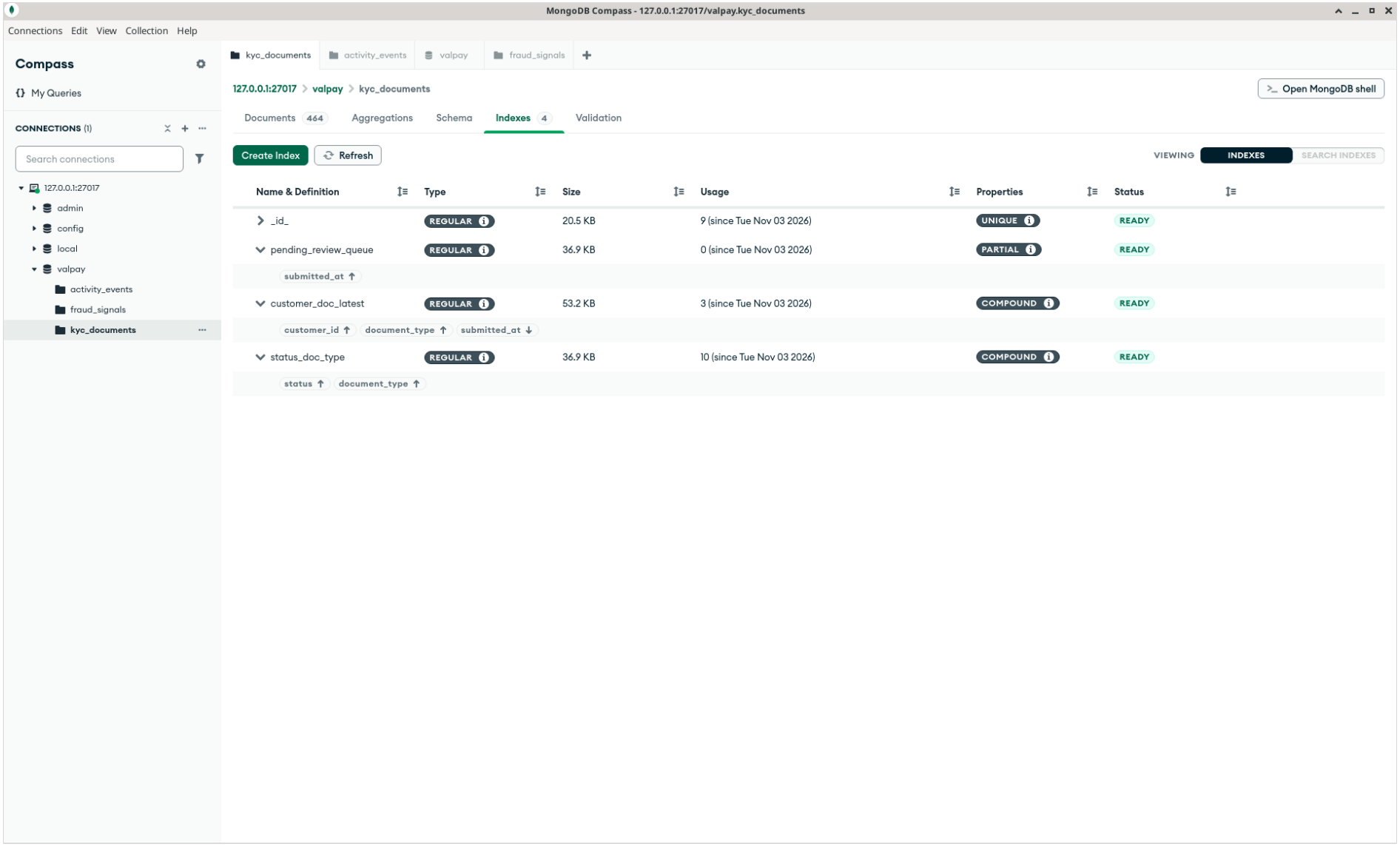


Figure 7. *MongoDB Compass: indexes on kyc_documents, including the partial pending_review_queue index.*

3. SCHEMA VALIDATION

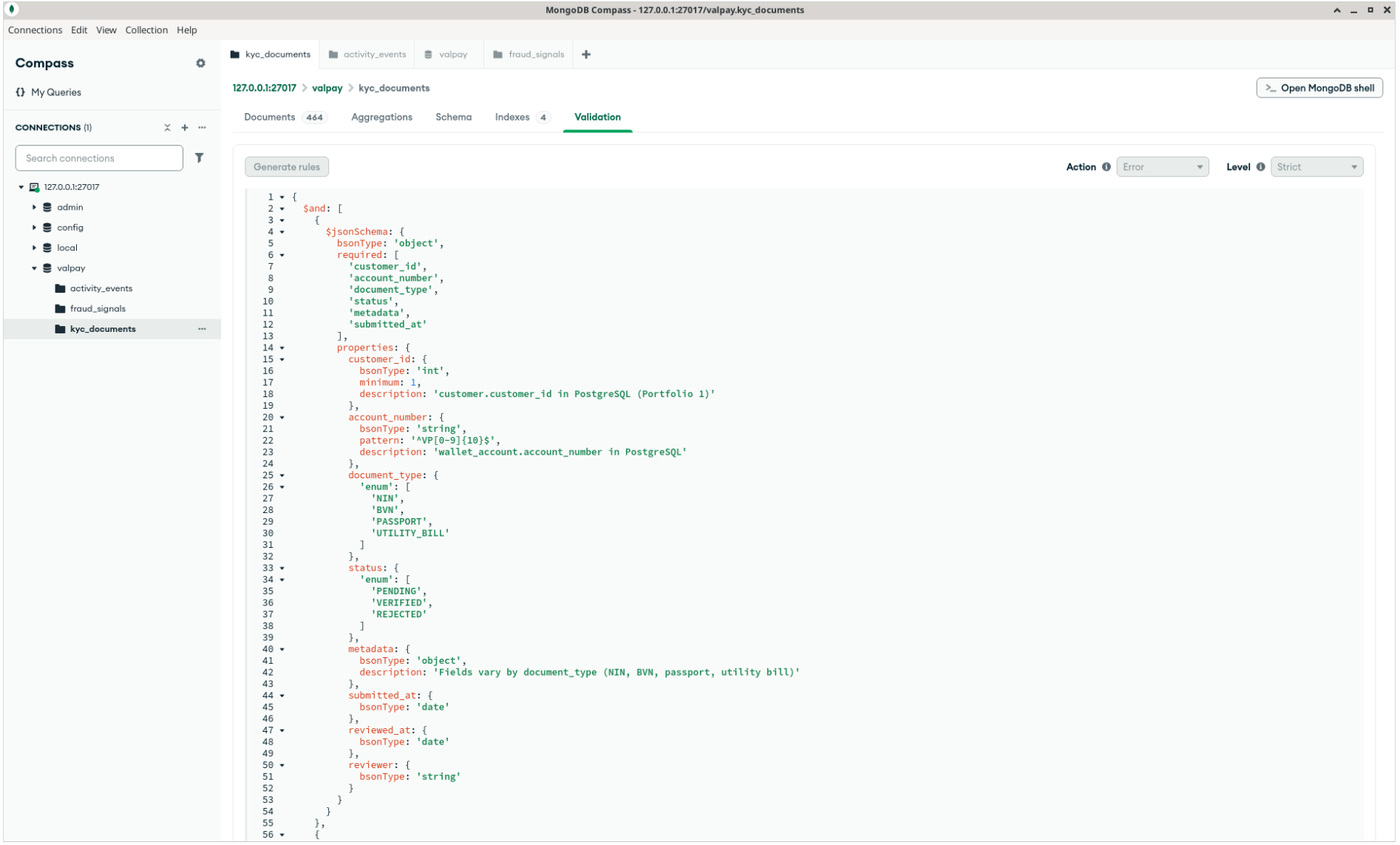


Figure 8. *MongoDB Compass: $jsonSchema validation rules on kyc_documents.*

```
portfolio-4-nosql $ mongosh --quiet valpay --file validation.mongodb.js | awk '/=== V2/{exit} {print}'
kyc_documents before tests: 464

=== V1. Unknown document type (DRIVERS_LICENCE) ===
Document: {"customer_id":57,"account_number":"VP2000000054","document_type":"DRIVERS_LICENCE","status":"PENDING",
    "metadata":{"masked_id":"A1234***","issuing_country":"NG"},"submitted_at":{"$date":"2026-11-01T09:00:00Z"}}
Result: REJECTED, code 121 (DocumentValidationFailure): Document failed validation
Why: {
  "operatorName": "$and",
  "clausesNotSatisfied": [
    {
      "index": 0,
      "details": {
        "operatorName": "$jsonSchema",
        "schemaRulesNotSatisfied": [
          {
            "operatorName": "properties",
            "propertiesNotSatisfied": [
              {
                "propertyName": "document_type",
                "details": [
                  {
                    "operatorName": "enum",
                    "specifiedAs": {
                      "enum": [
                        "NIN",
                        "BVN",
                        "PASSPORT",
                        "UTILITY_BILL"
                      ]
                    },
                    "reason": "value was not found in enum",
                    "consideredValue": "DRIVERS_LICENCE"
                  }
                ]
              }
            ]
          }
        ]
      }
    }
  ]
}

portfolio-4-nosql $
```

Figure 9. *mongosh: insert with an unknown document type rejected by the validator.*

```
portfolio-4-nosql $ mongosh --quiet valpay --file validation.mongodb.js | awk '/=== V4/,0'
=== V4. REJECTED without a rejection_reason ===
Document: {"customer_id":57,"account_number":"VP2000000054","document_type":"PASSPORT","status":"REJECTED",
    "metadata":{"masked_id":"A1234***","issuing_country":"NG"},"submitted_at":{"$date":"2026-11-01T09:00:00Z"},
    "reviewer":"compliance.agent01","reviewed_at":{"$date":"2026-11-01T12:00:00Z"}}
Result: REJECTED, code 121 (DocumentValidationFailure): Document failed validation
Why: {
  "operatorName": "$and",
  "clausesNotSatisfied": [
    {
      "index": 1,
      "details": {
        "operatorName": "$or",
        "clausesNotSatisfied": [
          {
            "index": 0,
            "details": {
              "operatorName": "$ne",
              "specifiedAs": {
                "status": {
                  "$ne": "REJECTED"
                }
              },
              "reason": "comparison succeeded",
              "consideredValue": "REJECTED"
            }
          },
          {
            "index": 1,
            "details": {
              "operatorName": "$type",
              "specifiedAs": {
                "metadata.rejection_reason": {
                  "$type": "string"
                }
              },
              "reason": "field was missing"
            }
          }
        ]
      }
    }
  ]
}

=== V5. Valid PASSPORT submission (accepted, then removed) ===
Document: {"customer_id":57,"account_number":"VP2000000054","document_type":"PASSPORT","status":"PENDING",
    "metadata":{"masked_id":"A1234***","issuing_country":"NG"},"submitted_at":{"$date":"2026-11-01T09:00:00Z"}}
Result: ACCEPTED

kyc_documents after tests: 464 (test document removed)
portfolio-4-nosql $
```

Figure 10. *mongosh: REJECTED document without rejection_reason refused, then a valid PASSPORT insert accepted.*

Source: `evidence/script-output/validation_rejection.txt`

```
kyc_documents before tests: 464

=== V1. Unknown document type (DRIVERS_LICENCE) ===
Document:
{"customer_id":57,"account_number":"VP2000000054","document_type":"DRIVERS_LICENCE","status":"PENDI
NG",
    "metadata":{"masked_id":"A1234***","issuing_country":"NG"},"submitted_at":
{"$date":"2026-11-01T09:00:00Z"}}
Result: REJECTED, code 121 (DocumentValidationFailure): Document failed validation
Why: {
  "operatorName": "$and",
  "clausesNotSatisfied": [
    {
      "index": 0,
      "details": {
        "operatorName": "$jsonSchema",
        "schemaRulesNotSatisfied": [
          {
            "operatorName": "properties",
            "propertiesNotSatisfied": [
```

```
              {
                "propertyName": "document_type",
                "details": [
                  {
                    "operatorName": "enum",
                    "specifiedAs": {
                      "enum": [
                        "NIN",
                        "BVN",
                        "PASSPORT",
                        "UTILITY_BILL"
                      ]
                    },
                    "reason": "value was not found in enum",
                    "consideredValue": "DRIVERS_LICENCE"
                  }
                ]
              }
            ]
          }
        ]
      }
    }
  ]
}
```

Source: evidence/script-output/validation_rejection.txt

```
=== V2. Old string customer id and a missing submitted_at ===
Document:
{"customer_id":"cust_valentine","account_number":"VP2000000054","document_type":"PASSPORT","status"
:"PENDING",
    "metadata":{"masked_id":"A1234***","issuing_country":"NG"}}
Result: REJECTED, code 121 (DocumentValidationFailure): Document failed validation
Why: {
  "operatorName": "$and",
  "clausesNotSatisfied": [
    {
      "index": 0,
      "details": {
        "operatorName": "$jsonSchema",
        "schemaRulesNotSatisfied": [
          {
            "operatorName": "properties",
            "propertiesNotSatisfied": [
              {
                "propertyName": "customer_id",
                "description": "customer.customer_id in PostgreSQL (Portfolio 1)",
                "details": [
                  {
                    "operatorName": "bsonType",
                    "specifiedAs": {
                      "bsonType": "int"
                    },
                    "reason": "type did not match",
                    "consideredValue": "cust_valentine",
                    "consideredType": "string"
                  }
                ]
              }
            ]
          },
          {
            "operatorName": "required",
            "specifiedAs": {
              "required": [
                "customer_id",
                "account_number",
```

```
                "document_type",
                "status",
                "metadata",
                "submitted_at"
              ]
            },
            "missingProperties": [
              "submitted_at"
            ]
          }
        ]
      }
    }
  ]
}
```

Source: evidence/script-output/validation_rejection.txt

```
=== V3. Account number not in the VP + 10 digits format ===
Document:
{"customer_id":57,"account_number":"0123456789","document_type":"PASSPORT","status":"PENDING",
    "metadata":{"masked_id":"A1234***","issuing_country":"NG"},"submitted_at":
{"$date":"2026-11-01T09:00:00Z"}}
Result: REJECTED, code 121 (DocumentValidationFailure): Document failed validation
Why: {
  "operatorName": "$and",
  "clausesNotSatisfied": [
    {
      "index": 0,
      "details": {
        "operatorName": "$jsonSchema",
        "schemaRulesNotSatisfied": [
          {
            "operatorName": "properties",
            "propertiesNotSatisfied": [
              {
                "propertyName": "account_number",
                "description": "wallet_account.account_number in PostgreSQL",
                "details": [
                  {
                    "operatorName": "pattern",
                    "specifiedAs": {
                      "pattern": "^VP[0-9]{10}$"
                    },
                    "reason": "regular expression did not match",
                    "consideredValue": "0123456789"
                  }
                ]
              }
            ]
          }
        ]
      }
    }
  ]
}
```

Source: evidence/script-output/validation_rejection.txt

```
=== V4. REJECTED without a rejection_reason ===
Document:
{"customer_id":57,"account_number":"VP2000000054","document_type":"PASSPORT","status":"REJECTED",
    "metadata":{"masked_id":"A1234***","issuing_country":"NG"},"submitted_at":
{"$date":"2026-11-01T09:00:00Z"},
    "reviewer":"compliance.agent01","reviewed_at":{"$date":"2026-11-01T12:00:00Z"}}
Result: REJECTED, code 121 (DocumentValidationFailure): Document failed validation
Why: {
```

```
  "operatorName": "$and",
  "clausesNotSatisfied": [
    {
      "index": 1,
      "details": {
        "operatorName": "$or",
        "clausesNotSatisfied": [
          {
            "index": 0,
            "details": {
              "operatorName": "$ne",
              "specifiedAs": {
                "status": {
                  "$ne": "REJECTED"
                }
              },
              "reason": "comparison succeeded",
              "consideredValue": "REJECTED"
            }
          },
          {
            "index": 1,
            "details": {
              "operatorName": "$type",
              "specifiedAs": {
                "metadata.rejection_reason": {
                  "$type": "string"
                }
              },
              "reason": "field was missing"
            }
          }
        ]
      }
    }
  ]
}
```


Source: `evidence/script-output/validation_rejection.txt`


```
=== V5. Valid PASSPORT submission (accepted, then removed) ===
Document: {"customer_id":57,"account_number":"VP2000000054","document_type":"PASSPORT","status":"PENDING",
    "metadata":{"masked_id":"A1234***","issuing_country":"NG"},"submitted_at":
{"$date":"2026-11-01T09:00:00Z"}}
Result: ACCEPTED

kyc_documents after tests: 464 (test document removed)
```

4. KYC QUERIES

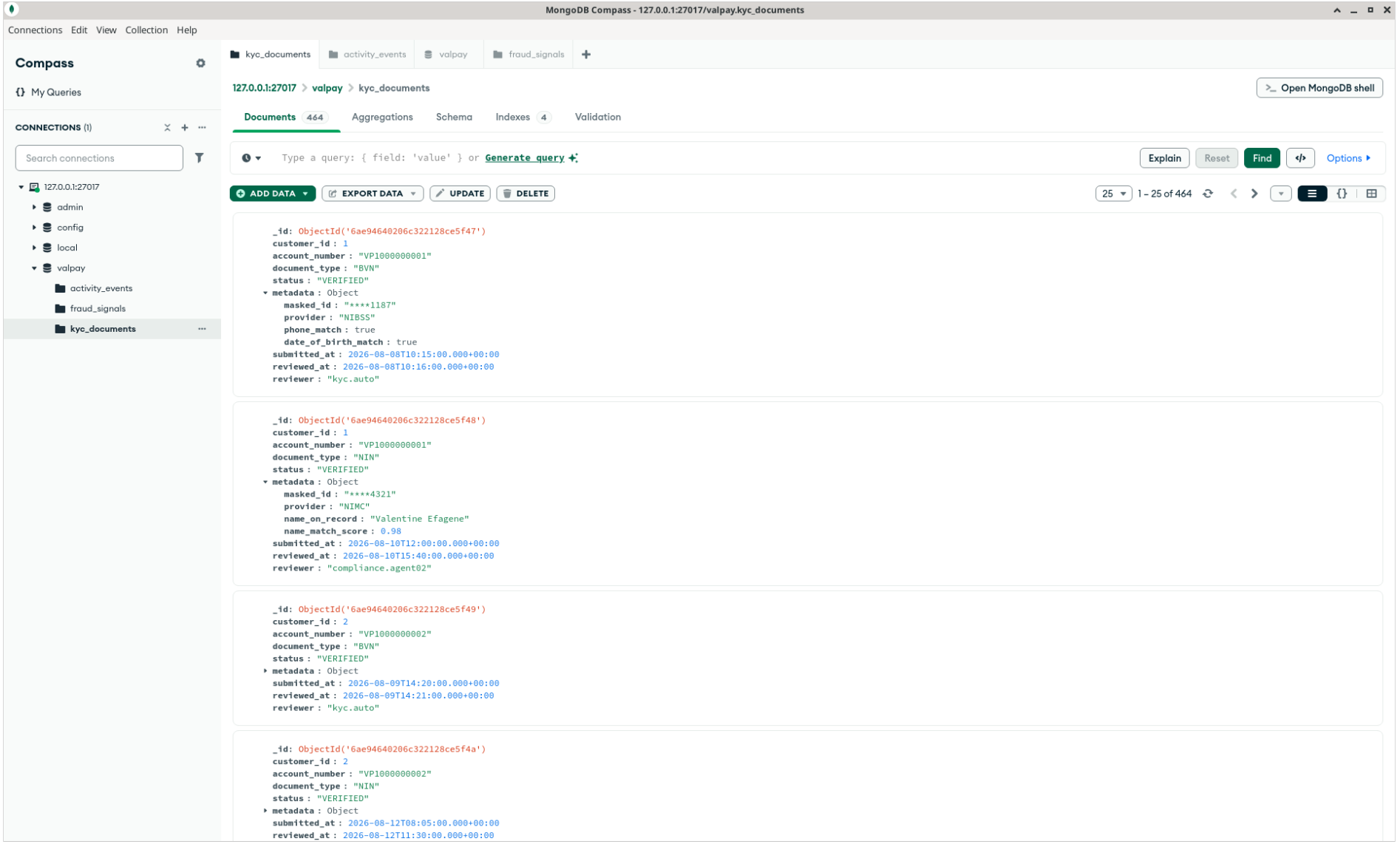


Figure 11. *MongoDB Compass: documents in kyc_documents.*

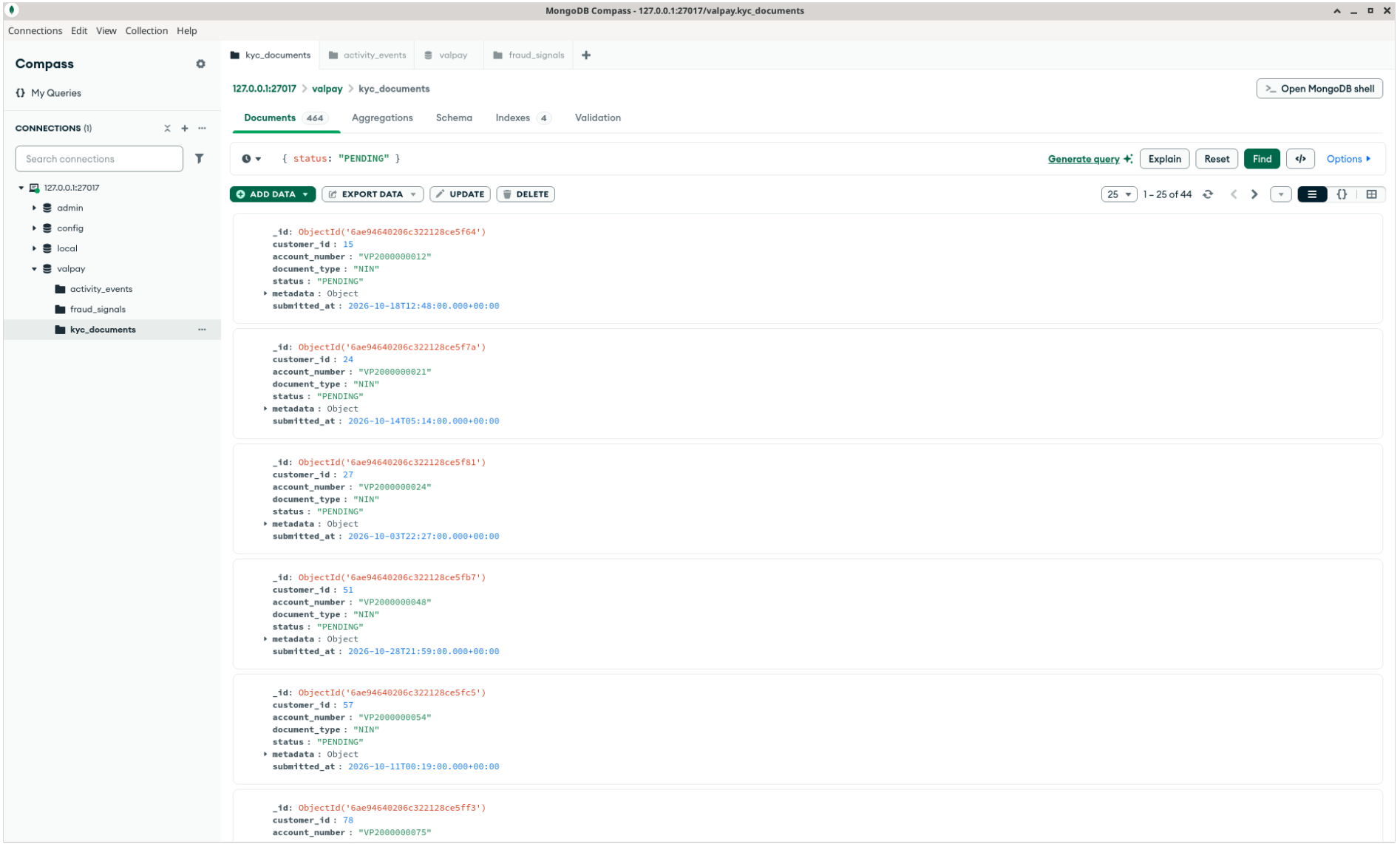


Figure 12. *MongoDB Compass: kyc_documents filtered on status PENDING.*

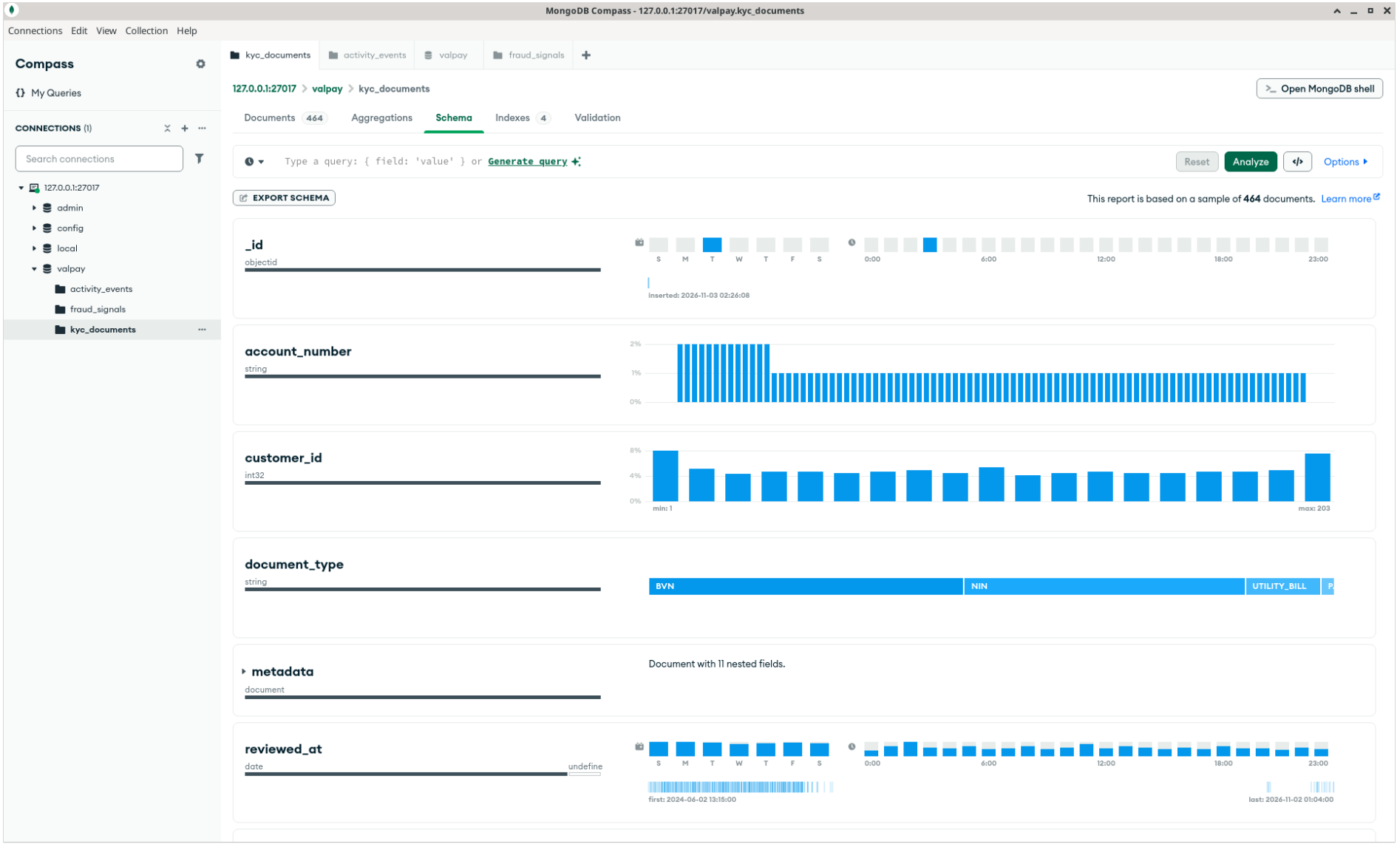


Figure 13. *MongoDB Compass: schema analysis of kyc_documents.*

```
portfolio-4-nosql $ mongosh --quiet valpay --file queries.mongodb.js | awk '/=== Q5/{exit} {print}'

=== Q1. KYC review queue (PENDING, oldest first, first 10) ===
Pending documents: 44
{"customer_id":139,"account_number":"VP2000000136","document_type":"UTILITY_BILL","metadata":{"provider":"PHED"},
    "submitted_at":{"$date":"2026-10-02T16:30:00Z"}}
{"customer_id":56,"account_number":"VP2000000053","document_type":"UTILITY_BILL","metadata":{"provider":"PHED"},"submitted_at":{"$date":"2026-10-02T17:10:00Z"}}
{"customer_id":27,"account_number":"VP2000000024","document_type":"NIN","metadata":{"provider":"NIMC"},"submitted_at":{"$date":"2026-10-03T22:27:00Z"}}
{"customer_id":70,"account_number":"VP2000000067","document_type":"UTILITY_BILL","metadata":{"provider":"AEDC"},"submitted_at":{"$date":"2026-10-05T12:13:00Z"}}
{"customer_id":141,"account_number":"VP2000000138","document_type":"NIN","metadata":{"provider":"NIMC"},"submitted_at":{"$date":"2026-10-09T02:07:00Z"}}
{"customer_id":85,"account_number":"VP2000000082","document_type":"UTILITY_BILL","metadata":{"provider":"IKEDC"},
    "submitted_at":{"$date":"2026-10-09T09:35:00Z"}}
{"customer_id":122,"account_number":"VP2000000119","document_type":"UTILITY_BILL","metadata":{"provider":"AEDC"},
    "submitted_at":{"$date":"2026-10-09T11:57:00Z"}}
{"customer_id":26,"account_number":"VP2000000023","document_type":"UTILITY_BILL","metadata":{"provider":"AEDC"},"submitted_at":{"$date":"2026-10-10T00:56:00Z"}}
{"customer_id":159,"account_number":"VP2000000156","document_type":"NIN","metadata":{"provider":"NIMC"},"submitted_at":{"$date":"2026-10-10T09:38:00Z"}}
{"customer_id":57,"account_number":"VP2000000054","document_type":"NIN","metadata":{"provider":"NIMC"},"submitted_at":{"$date":"2026-10-11T00:19:00Z"}}

=== Q2. KYC funnel by document type and status ===
{"total":202,"document_type":"BVN","verified":202,"pending":0,"rejected":0}
{"total":187,"document_type":"NIN","verified":142,"pending":15,"rejected":30}
{"total":63,"document_type":"UTILITY_BILL","verified":20,"pending":29,"rejected":14}
{"total":12,"document_type":"PASSPORT","verified":10,"pending":0,"rejected":2}

=== Q3. Rejection reasons by document type ===
{"n":12,"document_type":"NIN","reason":"nin_not_found"}
{"n":10,"document_type":"NIN","reason":"name_mismatch"}
{"n":8,"document_type":"NIN","reason":"blurry_image"}
{"n":8,"document_type":"UTILITY_BILL","reason":"address_mismatch"}
{"n":5,"document_type":"UTILITY_BILL","reason":"blurry_image"}
{"n":2,"document_type":"PASSPORT","reason":"expired_document"}
{"n":1,"document_type":"UTILITY_BILL","reason":"older_than_3_months"}

=== Q4. KYC tier derived from verified documents ===
{"customers":60,"tier":"TIER1"}
{"customers":122,"tier":"TIER2"}
{"customers":20,"tier":"TIER3"}

portfolio-4-nosql $
```

Figure 14. *mongosh: KYC review queue, funnel by document type and status, rejection reasons and derived tiers (Q1 to Q4).*

Source: `evidence/script-output/query_results.txt`


```
=== Q1. KYC review queue (PENDING, oldest first, first 10) ===
Pending documents: 44
{"customer_id":139,"account_number":"VP2000000136","document_type":"UTILITY_BILL","metadata":{"provider":"PHED"},
    "submitted_at":{"$date":"2026-10-02T16:30:00Z"}}
{"customer_id":56,"account_number":"VP2000000053","document_type":"UTILITY_BILL","metadata":{"provider":"PHED"},"submitted_at":{"$date":"2026-10-02T17:10:00Z"}}
{"customer_id":27,"account_number":"VP2000000024","document_type":"NIN","metadata":{"provider":"NIMC"},"submitted_at":{"$date":"2026-10-03T22:27:00Z"}}
{"customer_id":70,"account_number":"VP2000000067","document_type":"UTILITY_BILL","metadata":{"provider":"AEDC"},"submitted_at":{"$date":"2026-10-05T12:13:00Z"}}
{"customer_id":141,"account_number":"VP2000000138","document_type":"NIN","metadata":{"provider":"NIMC"},"submitted_at":{"$date":"2026-10-09T02:07:00Z"}}
```

```
{"customer_id":85,"account_number":"VP2000000082","document_type":"UTILITY_BILL","metadata":{"provider":"IKEDC"},
    "submitted_at":{"$date":"2026-10-09T09:35:00Z"}}
{"customer_id":122,"account_number":"VP2000000119","document_type":"UTILITY_BILL","metadata":{"provider":"AEDC"},
    "submitted_at":{"$date":"2026-10-09T11:57:00Z"}}
{"customer_id":26,"account_number":"VP2000000023","document_type":"UTILITY_BILL","metadata":{"provider":"AEDC"},"submitted_at":{"$date":"2026-10-10T00:56:00Z"}}
{"customer_id":159,"account_number":"VP2000000156","document_type":"NIN","metadata":{"provider":"NIMC"},"submitted_at":{"$date":"2026-10-10T09:38:00Z"}}
{"customer_id":57,"account_number":"VP2000000054","document_type":"NIN","metadata":{"provider":"NIMC"},"submitted_at":{"$date":"2026-10-11T00:19:00Z"}}
```

Source: `evidence/script-output/query_results.txt`

```
=== Q2. KYC funnel by document type and status ===
{"total":202,"document_type":"BVN","verified":202,"pending":0,"rejected":0}
{"total":187,"document_type":"NIN","verified":142,"pending":15,"rejected":30}
{"total":63,"document_type":"UTILITY_BILL","verified":20,"pending":29,"rejected":14}
{"total":12,"document_type":"PASSPORT","verified":10,"pending":0,"rejected":2}
```

Source: `evidence/script-output/query_results.txt`

```
=== Q3. Rejection reasons by document type ===
{"n":12,"document_type":"NIN","reason":"nin_not_found"}
{"n":10,"document_type":"NIN","reason":"name_mismatch"}
{"n":8,"document_type":"NIN","reason":"blurry_image"}
{"n":8,"document_type":"UTILITY_BILL","reason":"address_mismatch"}
{"n":5,"document_type":"UTILITY_BILL","reason":"blurry_image"}
{"n":2,"document_type":"PASSPORT","reason":"expired_document"}
{"n":1,"document_type":"UTILITY_BILL","reason":"older_than_3_months"}
```

Source: `evidence/script-output/query_results.txt`

```
=== Q4. KYC tier derived from verified documents ===
{"customers":60,"tier":"TIER1"}
{"customers":122,"tier":"TIER2"}
{"customers":20,"tier":"TIER3"}
```

5. ACTIVITY QUERIES

```
portfolio-4-nosql $ mongosh --quiet valpay --file queries.mongodb.js | awk '/=== Q5/,/=== Q7/; /=== Q11/,/=== Q12/' | grep -v '=== Q7\|=== Q12'
=== Q5. Customer timeline, customer_id 57 (latest 12 events) ===
{"event":"transfer_initiated","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","reference":"VP-TXN-20261026-039684","amount_ngn":18070.29,
    "ts":{"$date":"2026-10-26T16:51:58Z"}}
{"event":"otp_sent","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-26T16:50:58Z"}}
{"event":"login_success","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-26T16:48:58Z"}}
{"event":"transfer_initiated","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","reference":"VP-TXN-20261025-039625","amount_ngn":25000.76,
    "ts":{"$date":"2026-10-25T17:23:59Z"}}
{"event":"otp_sent","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-25T17:22:59Z"}}
{"event":"login_success","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-25T17:20:59Z"}}
{"event":"transfer_initiated","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","reference":"VP-TXN-20261023-039512","amount_ngn":9490.26,
    "ts":{"$date":"2026-10-23T19:48:23Z"}}
{"event":"otp_sent","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-23T19:47:23Z"}}
{"event":"login_success","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-23T19:45:23Z"}}
{"event":"transfer_initiated","device_id":"dev_14bacc0e9a20","ip":"105.112.234.201","reference":"VP-TXN-20261020-039301","amount_ngn":20150.5,
    "ts":{"$date":"2026-10-20T07:18:11Z"}}
{"event":"otp_sent","device_id":"dev_14bacc0e9a20","ip":"105.112.234.201","ts":{"$date":"2026-10-20T07:17:11Z"}}
{"event":"login_success","device_id":"dev_14bacc0e9a20","ip":"105.112.234.201","ts":{"$date":"2026-10-20T07:15:11Z"}}

=== Q6. Transfers initiated 1-7 October 2026 ===
{"transfers":291,"txn_status":"SUCCESS","total_ngn":4365034.68}
{"transfers":2,"txn_status":"PENDING","total_ngn":37821}
Largest five in the window:
{"account_number":"VP2000000144","reference":"VP-TXN-20261001-038160","amount_ngn":29920.78,"beneficiary":"VP2000000098","ts":{"$date":"2026-10-01T05:05:21Z"}}
{"account_number":"VP2000000091","reference":"VP-TXN-20261003-038321","amount_ngn":29910.11,"beneficiary":"VP2000000192","ts":{"$date":"2026-10-03T21:52:11Z"}}
{"account_number":"VP2000000026","reference":"VP-TXN-20261001-038152","amount_ngn":29820.12,"beneficiary":"VP2000000199","ts":{"$date":"2026-10-01T01:56:49Z"}}
{"account_number":"VP2000000190","reference":"VP-TXN-20261005-038442","amount_ngn":29740.9,"beneficiary":"VP2000000042","ts":{"$date":"2026-10-05T22:16:20Z"}}
{"account_number":"VP2000000041","reference":"VP-TXN-20261003-038283","amount_ngn":29690.38,"beneficiary":"VP2000000137","ts":{"$date":"2026-10-03T06:41:38Z"}}

=== Q11. Trace VP-TXN-OK-004 (the Portfolio 3 transfer) and the session around it ===
{"event":"transfer_initiated","customer_id":1,"account_number":"VP1000000001","device_id":"dev_e94bebb5927b","ip":"105.112.129.46","channel":"ios",
    "reference":"VP-TXN-OK-004","amount_ngn":2500,"beneficiary":"VP1000000002","txn_status":"SUCCESS","ts":{"$date":"2026-10-28T07:05:00Z"}}
{"event":"login_success","device_id":"dev_e94bebb5927b","ip":"105.112.129.46","ts":{"$date":"2026-10-28T07:02:00Z"}}
{"event":"otp_sent","device_id":"dev_e94bebb5927b","ip":"105.112.129.46","ts":{"$date":"2026-10-28T07:04:00Z"}}
{"event":"transfer_initiated","device_id":"dev_e94bebb5927b","ip":"105.112.129.46","ts":{"$date":"2026-10-28T07:05:00Z"}}

portfolio-4-nosql $
```

Figure 15. *mongosh: customer timeline, transfers initiated 1 to 7 October 2026 and the VP-TXN-OK-004 trace (Q5, Q6, Q11).*

Source: `evidence/script-output/query_results.txt`


```
=== Q5. Customer timeline, customer_id 57 (latest 12 events) ===
{"event":"transfer_initiated","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","reference":"VP-TXN-20261026-039684","amount_ngn":18070.29,
    "ts":{"$date":"2026-10-26T16:51:58Z"}}
{"event":"otp_sent","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-26T16:50:58Z"}}
{"event":"login_success","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-26T16:48:58Z"}}
{"event":"transfer_initiated","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","reference":"VP-TXN-20261025-039625","amount_ngn":25000.76,
```

```
    "ts":{"$date":"2026-10-25T17:23:59Z"}}
{"event":"otp_sent","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-25T17:22:59Z"}}
{"event":"login_success","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-25T17:20:59Z"}}
{"event":"transfer_initiated","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","reference":"VP-TXN-20261023-039512","amount_ngn":9490.26,
    "ts":{"$date":"2026-10-23T19:48:23Z"}}
{"event":"otp_sent","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-23T19:47:23Z"}}
{"event":"login_success","device_id":"dev_aaa1d3733c62","ip":"105.112.234.201","ts":{"$date":"2026-10-23T19:45:23Z"}}
{"event":"transfer_initiated","device_id":"dev_14bacc0e9a20","ip":"105.112.234.201","reference":"VP-TXN-20261020-039301","amount_ngn":20150.5,
    "ts":{"$date":"2026-10-20T07:18:11Z"}}
{"event":"otp_sent","device_id":"dev_14bacc0e9a20","ip":"105.112.234.201","ts":{"$date":"2026-10-20T07:17:11Z"}}
{"event":"login_success","device_id":"dev_14bacc0e9a20","ip":"105.112.234.201","ts":{"$date":"2026-10-20T07:15:11Z"}}
```


Source: `evidence/script-output/query_results.txt`


```
=== Q6. Transfers initiated 1-7 October 2026 ===
{"transfers":291,"txn_status":"SUCCESS","total_ngn":4365034.68}
{"transfers":2,"txn_status":"PENDING","total_ngn":37821}
Largest five in the window:
{"account_number":"VP2000000144","reference":"VP-TXN-20261001-038160","amount_ngn":29920.78,"beneficiary":"VP2000000098","ts":{"$date":"2026-10-01T05:05:21Z"}}
{"account_number":"VP2000000091","reference":"VP-TXN-20261003-038321","amount_ngn":29910.11,"beneficiary":"VP2000000192","ts":{"$date":"2026-10-03T21:52:11Z"}}
{"account_number":"VP2000000026","reference":"VP-TXN-20261001-038152","amount_ngn":29820.12,"beneficiary":"VP2000000199","ts":{"$date":"2026-10-01T01:56:49Z"}}
{"account_number":"VP2000000190","reference":"VP-TXN-20261005-038442","amount_ngn":29740.9,"beneficiary":"VP2000000042","ts":{"$date":"2026-10-05T22:16:20Z"}}
{"account_number":"VP2000000041","reference":"VP-TXN-20261003-038283","amount_ngn":29690.38,"beneficiary":"VP2000000137","ts":{"$date":"2026-10-03T06:41:38Z"}}
```


Source: `evidence/script-output/query_results.txt`


```
=== Q11. Trace VP-TXN-OK-004 (the Portfolio 3 transfer) and the session around it ===
{"event":"transfer_initiated","customer_id":1,"account_number":"VP1000000001","device_id":"dev_e94bebb5927b","ip":"105.112.129.46","channel":"ios",
    "reference":"VP-TXN-OK-004","amount_ngn":2500,"beneficiary":"VP1000000002","txn_status":"SUCCESS","ts":{"$date":"2026-10-28T07:05:00Z"}}
{"event":"login_success","device_id":"dev_e94bebb5927b","ip":"105.112.129.46","ts":{"$date":"2026-10-28T07:02:00Z"}}
{"event":"otp_sent","device_id":"dev_e94bebb5927b","ip":"105.112.129.46","ts":{"$date":"2026-10-28T07:04:00Z"}}
{"event":"transfer_initiated","device_id":"dev_e94bebb5927b","ip":"105.112.129.46","ts":{"$date":"2026-10-28T07:05:00Z"}}
```

6. FRAUD QUERIES

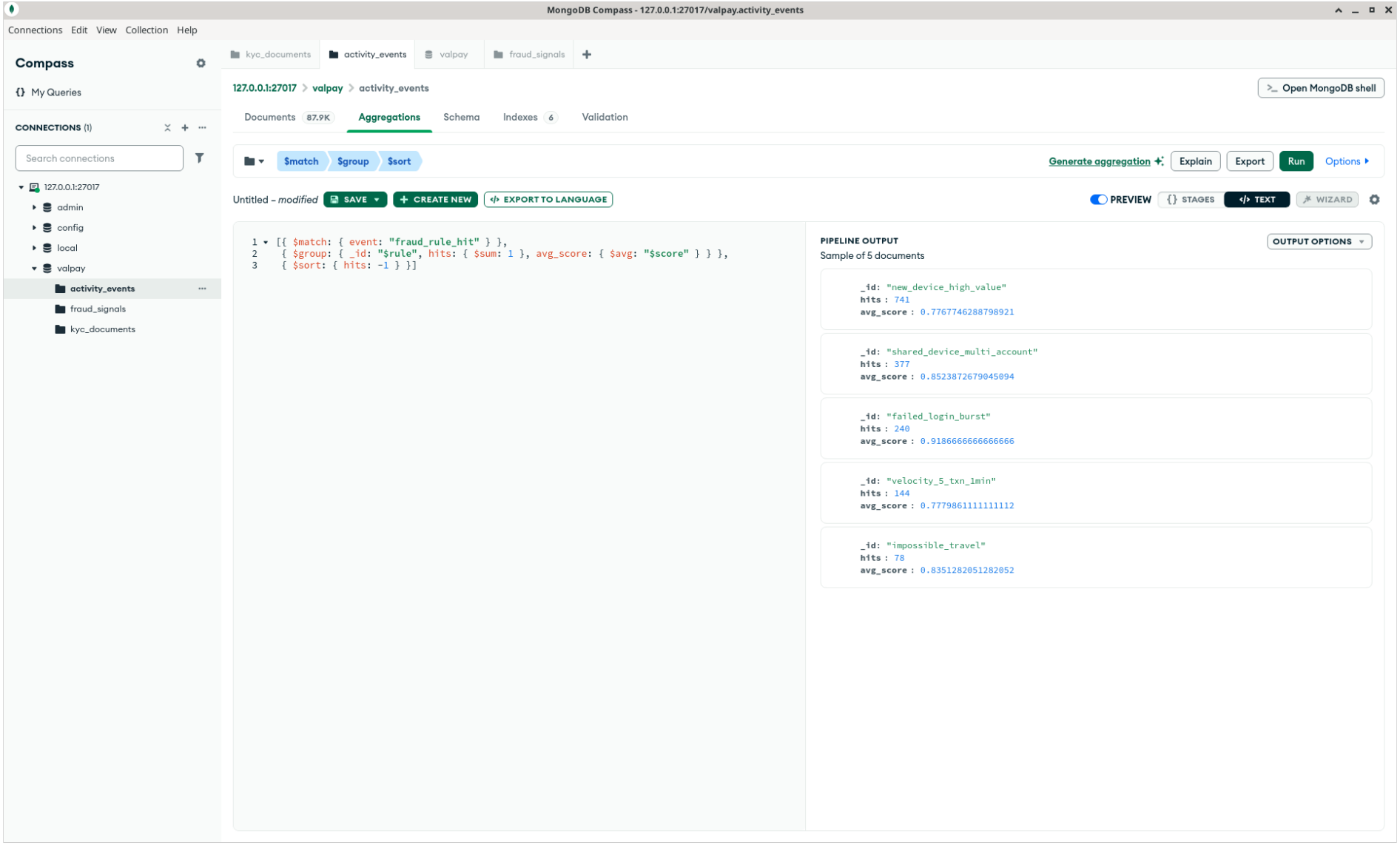


Figure 16. *MongoDB Compass: aggregation of fraud_rule_hit events grouped by rule, with hits and average score.*

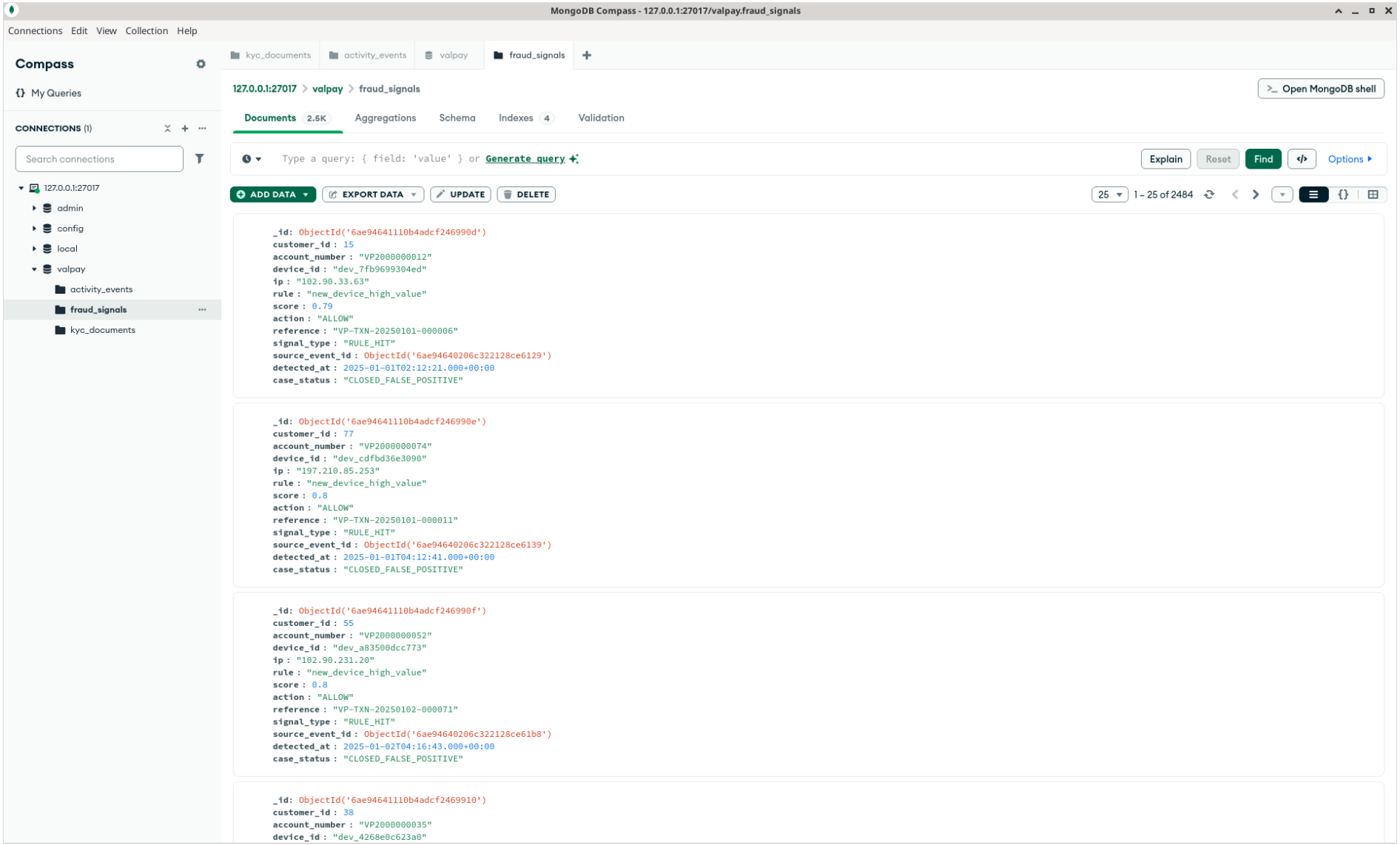


Figure 17. *MongoDB Compass: documents in fraud_signals.*

```
portfolio-4-nosql $ mongosh --quiet valpay --file queries.mongodb.js | awk '/=== Q7/,/=== Q10/' | grep -v '=== Q10'
=== Q7. Fraud rule hits by rule ===
{"hits":741,"max_score":0.99,"step_up":259,"blocked":0,"rule":"new_device_high_value","customers":195,"avg_score":0.777}
{"hits":377,"max_score":0.99,"step_up":377,"blocked":0,"rule":"shared_device_multi_account","customers":40,"avg_score":0.852}
{"hits":240,"max_score":0.99,"step_up":0,"blocked":240,"rule":"failed_login_burst","customers":135,"avg_score":0.919}
{"hits":144,"max_score":0.99,"step_up":144,"blocked":0,"rule":"velocity_5_txn_1min","customers":109,"avg_score":0.778}
{"hits":78,"max_score":0.99,"step_up":78,"blocked":0,"rule":"impossible_travel","customers":62,"avg_score":0.835}

=== Q8. Fraud rule hits per month, 2026 ===
{"hits":65,"month":"2026-01"}
{"hits":73,"month":"2026-02"}
{"hits":76,"month":"2026-03"}
{"hits":75,"month":"2026-04"}
{"hits":81,"month":"2026-05"}
{"hits":72,"month":"2026-06"}
{"hits":75,"month":"2026-07"}
{"hits":66,"month":"2026-08"}
{"hits":68,"month":"2026-09"}
{"hits":77,"month":"2026-10"}

=== Q9. Top 8 risky devices ===
{"failed_logins":246,"rule_hits":135,"last_seen":{"$date":"2026-10-21T14:03:00Z"},"device_id":"dev_4268e0c623a0","customer_count":49,"ip_count":1,"risk_score":0.99}
{"failed_logins":245,"rule_hits":132,"last_seen":{"$date":"2026-10-29T17:04:07Z"},"device_id":"dev_84f435b57fbd","customer_count":51,"ip_count":1,"risk_score":0.99}
{"failed_logins":236,"rule_hits":123,"last_seen":{"$date":"2026-10-24T23:01:00Z"},"device_id":"dev_899a0b812d62","customer_count":49,"ip_count":1,"risk_score":0.99}
{"failed_logins":241,"rule_hits":123,"last_seen":{"$date":"2026-10-22T18:52:15Z"},"device_id":"dev_ae0269b0efeb","customer_count":49,"ip_count":1,"risk_score":0.99}
{"failed_logins":248,"rule_hits":106,"last_seen":{"$date":"2026-10-27T08:30:15Z"},"device_id":"dev_f4c0bf3dae82","customer_count":50,"ip_count":1,"risk_score":0.99}
{"failed_logins":2,"rule_hits":4,"last_seen":{"$date":"2026-10-31T08:38:41Z"},"device_id":"dev_767d176b410e","customer_count":1,"ip_count":5,"risk_score":0.51}
{"failed_logins":5,"rule_hits":3,"last_seen":{"$date":"2026-10-18T09:10:26Z"},"device_id":"dev_2af25aa3f906","customer_count":1,"ip_count":2,"risk_score":0.51}
{"failed_logins":3,"rule_hits":2,"last_seen":{"$date":"2026-10-21T07:38:17Z"},"device_id":"dev_3f42b817d261","customer_count":1,"ip_count":2,"risk_score":0.51}

portfolio-4-nosql $
```

Figure 18. *mongosh: fraud rule hits by rule and the top risky devices.*

Source: `evidence/script-output/query_results.txt`


```
=== Q7. Fraud rule hits by rule ===
{"hits":741,"max_score":0.99,"step_up":259,"blocked":0,"rule":"new_device_high_value","customers":195,"avg_score":0.777}
{"hits":377,"max_score":0.99,"step_up":377,"blocked":0,"rule":"shared_device_multi_account","customers":40,"avg_score":0.852}
{"hits":240,"max_score":0.99,"step_up":0,"blocked":240,"rule":"failed_login_burst","customers":135,"avg_score":0.919}
{"hits":144,"max_score":0.99,"step_up":144,"blocked":0,"rule":"velocity_5_txn_1min","customers":109,"avg_score":0.778}
{"hits":78,"max_score":0.99,"step_up":78,"blocked":0,"rule":"impossible_travel","customers":62,"avg_score":0.835}
```


Source: `evidence/script-output/query_results.txt`


```
=== Q8. Fraud rule hits per month, 2026 ===
{"hits":65,"month":"2026-01"}
{"hits":73,"month":"2026-02"}
{"hits":76,"month":"2026-03"}
{"hits":75,"month":"2026-04"}
{"hits":81,"month":"2026-05"}
```

```
{"hits":72,"month":"2026-06"}
{"hits":75,"month":"2026-07"}
{"hits":66,"month":"2026-08"}
{"hits":68,"month":"2026-09"}
{"hits":77,"month":"2026-10"}
```

Source: evidence/script-output/query_results.txt

```
=== Q9. Top 8 risky devices ===
{"failed_logins":246,"rule_hits":135,"last_seen":{"$date":"2026-10-21T14:03:00Z"},"device_id":"dev_4268e0c623a0","customer_count":49,"ip_count":1,
    "risk_score":0.99}
{"failed_logins":245,"rule_hits":132,"last_seen":{"$date":"2026-10-29T17:04:07Z"},"device_id":"dev_84f435b57fbd","customer_count":51,"ip_count":1,
    "risk_score":0.99}
{"failed_logins":236,"rule_hits":123,"last_seen":{"$date":"2026-10-24T23:01:00Z"},"device_id":"dev_899a0b812d62","customer_count":49,"ip_count":1,
    "risk_score":0.99}
{"failed_logins":241,"rule_hits":123,"last_seen":{"$date":"2026-10-22T18:52:15Z"},"device_id":"dev_ae0269b0efeb","customer_count":49,"ip_count":1,
    "risk_score":0.99}
{"failed_logins":248,"rule_hits":106,"last_seen":{"$date":"2026-10-27T08:30:15Z"},"device_id":"dev_f4c0bf3dae82","customer_count":50,"ip_count":1,
    "risk_score":0.99}
{"failed_logins":2,"rule_hits":4,"last_seen":{"$date":"2026-10-31T08:38:41Z"},"device_id":"dev_767d176b410e","customer_count":1,"ip_count":5,"risk_score":0.51}
{"failed_logins":5,"rule_hits":3,"last_seen":{"$date":"2026-10-18T09:10:26Z"},"device_id":"dev_2af25aa3f906","customer_count":1,"ip_count":2,"risk_score":0.51}
{"failed_logins":3,"rule_hits":2,"last_seen":{"$date":"2026-10-21T07:38:17Z"},"device_id":"dev_3f42b817d261","customer_count":1,"ip_count":2,"risk_score":0.51}
```

Source: evidence/script-output/query_results.txt

```
=== Q10. Open fraud cases (newest first, first 8) ===
Open cases: 58
{"customer_id":73,"rule":"velocity_5_txn_1min","score":0.98,"action":"STEP_UP","reference":"VP-TXN-20261031-039964",
    "detected_at":{"$date":"2026-10-31T09:10:52Z"}}
{"customer_id":59,"rule":"new_device_high_value","score":0.6,"action":"ALLOW","reference":"VP-TXN-20261031-039960",
    "detected_at":{"$date":"2026-10-31T07:46:35Z"}}
{"customer_id":142,"rule":"new_device_high_value","score":0.95,"action":"STEP_UP","reference":"VP-TXN-20261031-039952",
    "detected_at":{"$date":"2026-10-31T04:38:03Z"}}
{"customer_id":159,"rule":"velocity_5_txn_1min","score":0.61,"action":"STEP_UP","reference":"VP-TXN-20261030-039910",
    "detected_at":{"$date":"2026-10-30T11:43:14Z"}}
{"customer_id":9,"rule":"new_device_high_value","score":0.67,"action":"STEP_UP","reference":"VP-TXN-20261030-039905",
    "detected_at":{"$date":"2026-10-30T09:42:53Z"}}
{"customer_id":63,"rule":"shared_device_multi_account","score":0.85,"action":"STEP_UP","reference":"VP-TXN-20261029-039864",
    "detected_at":{"$date":"2026-10-29T17:04:07Z"}}
{"customer_id":144,"rule":"new_device_high_value","score":0.59,"action":"ALLOW","reference":"VP-TXN-20261028-039790",
    "detected_at":{"$date":"2026-10-28T11:35:08Z"}}
{"customer_id":38,"rule":"new_device_high_value","score":0.58,"action":"STEP_UP","reference":"VP-TXN-20261028-039766",
    "detected_at":{"$date":"2026-10-28T01:49:31Z"}}
```

Source: evidence/script-output/query_results.txt

```
=== Q12. Customers on shared-device rule hits with KYC still pending ===
{"hits":11,"last_hit":{"$date":"2026-07-07T22:54:07Z"},"pending_kyc":[{"document_type":"UTILITY_BILL","submitted_at":{"$date":"2026-10-25T03:29:00Z"}}],
    "customer_id":167}
{"hits":10,"last_hit":{"$date":"2026-09-17T21:19:43Z"},"pending_kyc":[{"document_type":"UTILITY_BILL","submitted_at":{"$date":"2026-10-25T10:48:00Z"}}],
    "customer_id":67}
{"hits":10,"last_hit":{"$date":"2026-10-02T05:09:25Z"},"pending_kyc":[{"document_type":"NIN","submitted_at":{"$date":"2026-10-10T09:38:00Z"}}],
    "customer_id":159}
{"hits":10,"last_hit":{"$date":"2026-10-24T23:01:00Z"},"pending_kyc":[{"document_type":"UTILITY_BILL","submitted_at":{"$date":"2026-10-16T22:11:00Z"}}],
    "customer_id":50}
{"hits":9,"last_hit":{"$date":"2026-07-27T02:31:13Z"},"pending_kyc":[{"document_type":"UTILITY_BILL","submitted_at":{"$date":"2026-10-25T11:39:00Z"}}],
    "customer_id":200}
{"hits":9,"last_hit":{"$date":"2026-08-11T12:09:14Z"},"pending_kyc":[{"document_type":"NIN","submitted_at":
{"$date":"2026-10-17T07:34:00Z"}}],"customer_id":192}
{"hits":8,"last_hit":{"$date":"2026-08-01T19:10:10Z"},"pending_kyc":[{"document_type":"UTILITY_BILL","submitted_at":{"$date":"2026-10-19T04:00:00Z"}}],
    "customer_id":134}
```

7. CROSS-STORE CHECK WITH POSTGRESQL

```
advanced-database-portfolio $ MONGO_MODE=local scripts/nosql_evidence.sh && cat portfolio-4-nosql/evidence/cross_store_check.txt
Waiting for MongoDB...
Loading seed...
Running queries...
Running explain before/after...
Running validation tests...
Refreshing sample_kyc.jsonl and sample_activity.jsonl...
Cross-checking against PostgreSQL...
Done. Portfolio 4 evidence written to portfolio-4-nosql/evidence/
=== Cross-store check: MongoDB (valpay) against PostgreSQL (valpay) ===

Transfers: transfer_initiated events vs wallet_transaction + transfer_detail
  (reference | amount | from account | to account)
  MongoDB events            : 28003
  PostgreSQL transfers      : 28003
  identical in both         : 28003
  only in MongoDB           : 0
  only in PostgreSQL        : 0

KYC tiers: tier derived from VERIFIED kyc_documents vs customer.kyc_tier (ops wallet 3 excluded)
  customers compared        : 202
  same tier                 : 202
  different or missing      : 0

Sample: VP-TXN-OK-004 in both stores
  MongoDB   : VP-TXN-OK-004|2500.00|VP1000000001|VP1000000002
  PostgreSQL: VP-TXN-OK-004|2500.00|VP1000000001|VP1000000002
advanced-database-portfolio $
```

Figure 19. *Evidence runner output and the cross-store check between MongoDB and PostgreSQL.*

Source: `evidence/script-output/cross_store_check.txt`


```
=== Cross-store check: MongoDB (valpay) against PostgreSQL (valpay) ===

Transfers: transfer_initiated events vs wallet_transaction + transfer_detail
  (reference | amount | from account | to account)
  MongoDB events            : 28003
  PostgreSQL transfers      : 28003
  identical in both         : 28003
  only in MongoDB           : 0
  only in PostgreSQL        : 0

KYC tiers: tier derived from VERIFIED kyc_documents vs customer.kyc_tier (ops wallet 3 excluded)
  customers compared        : 202
  same tier                 : 202
  different or missing      : 0

Sample: VP-TXN-OK-004 in both stores
  MongoDB   : VP-TXN-OK-004|2500.00|VP1000000001|VP1000000002
  PostgreSQL: VP-TXN-OK-004|2500.00|VP1000000001|VP1000000002
```